Dual Nature of Radiation and Matter

Complete Conceptual Guide, Derivations, Graphical Analysis & NEET Question Bank

1. Syllabus Overview & Learning Objectives

In NEET UG, the **Dual Nature of Radiation and Matter** is one of the highest-yield chapters under Modern Physics. It accounts for 2 to 3 direct questions every year. Questions systematically test the graphical interpretations of the photoelectric effect, stopping potential calculations, photon momentum, energy quantization, and de Broglie matter waves.

Core Learning Goals for NEET UG:

- Understand historical cathode ray experiments and the universality of the electron's specific charge (e/m).
- Distinguish between thermionic, field, and photoelectric emission mechanisms.
- Master Hallwachs' and Lenard's observations and the experimental apparatus of the photoelectric effect.
- Comprehend why classical electromagnetic wave theory failed to explain photoelectric phenomena.
- Apply Einstein's photoelectric equation to calculate work function (ϕ_0), stopping potential (V_0), maximum kinetic energy ($K_{\max}$), and threshold frequency (ν_0).
- Interpret and manipulate all four standard photoelectric graphs (I vs. Intensity, I vs. Potential, and V_0 vs. ν).
- Understand the photon picture of electromagnetic radiation (energy, momentum, and neutrality).
- Calculate the de Broglie wavelength (λ) for both sub-atomic particles and macroscopic bodies.

BOSONANCE

2. Historical Foundations & Electron Discovery

By the late 19th century, Maxwell's equations and Heinrich Hertz's experiments (1887) on the generation and detection of electromagnetic waves had firmly established the *wave nature of light*. However, electrical discharge studies in gases at low pressure (around 0.001 mm of mercury column) revealed unexpected phenomena:

- **Cathode Rays:** Observed by William Crookes in 1870, who in 1879 suggested they were streams of fast-moving negatively charged particles. When cathode rays strike the soda glass opposite the cathode, a characteristic yellowish-green fluorescent glow appears.
- **Universality of Electron (e/m):** In 1897, J. J. Thomson determined the speed and specific charge (charge-to-mass ratio, e/m) of cathode ray particles using mutually perpendicular electric and magnetic fields (velocity selector principle). The particles moved at speeds ranging from $0.1c$ to $0.2c$. The currently accepted value is:

$$\frac{e}{m} = 1.76 \times 10^{11}\ \text{C/kg}$$

  Crucially, this ratio was completely independent of the metal used as the cathode or the nature of the gas inside the discharge tube, proving that these negatively charged corpuscles—named **electrons**—are universal fundamental constituents of matter.
- **Quantization of Charge:** In 1913, R. A. Millikan's oil-drop experiment demonstrated that electric charge always occurs in integral multiples of a basic elementary quantum:

$$q = ne, \quad \text{where } e = 1.602 \times 10^{-19}\ \text{C} \quad (n = \pm 1, \pm 2, \ldots)$$

Combining e and e/m allowed the precise calculation of electron rest mass $m = 9.11 \times 10^{-31}\ \text{kg}$.

3. Electron Emission and Work Function

Although free electrons inside a conductor move freely among lattice ions, they cannot spontaneously leave the metal surface. When an electron attempts to pull away, the surface immediately acquires a net positive charge that exerts a strong electrostatic restoring force. Hence, an electron must absorb an external threshold energy to escape.

Work Function (ϕ_0): The minimum energy required by an electron to escape from a metal surface.

It is a characteristic property of the given metal and the physical condition of its surface. It is conventionally measured in **electron volts (eV)**:

$$1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$$

Note: Caesium has one of the lowest work functions ($\approx 2.14\ \text{eV}$), making it sensitive even to visible light, whereas platinum has a high work function ($\approx 5.65\ \text{eV}$).

Physical Mechanisms of Electron Emission

Process	Energy Mechanism	Typical Application / Condition
Thermionic Emission	Thermal energy supplied by heating allows electrons to overcome ϕ_0.	Vacuum tubes, cathode-ray filament guns.
Field Emission (Cold Cathode)	Subjecting the metal to a very strong electric field ($\sim 10^8\ \text{V/m}$) pulls electrons out.	Spark plugs, field-emission microscopes.
Photoelectric Emission	Incident radiation of suitable frequency illuminates the surface; energy is absorbed by electrons.	Photocells, solar detectors, radiation sensors.

4. Experimental Investigation of the Photoelectric Effect

In 1887, Heinrich Hertz noticed that electric sparks jumped more readily across detector gaps when illuminated by ultraviolet light from an arc lamp. Wilhelm Hallwachs (1888) and Philipp Lenard (1900–1902) followed up with systematic studies:

- An uncharged zinc plate connected to an electroscope became positively charged under UV irradiation.
- A negatively charged zinc plate lost its charge rapidly under UV light.
- A positively charged zinc plate had its positive charge further enhanced by UV light.

These observations proved that negative charges (electrons) were emitted from the metal surface upon absorbing light above a critical **threshold frequency (ν_0)**. Certain metals such as zinc, cadmium, and magnesium respond only to UV radiation, whereas alkali metals (lithium, sodium, potassium, rubidium, caesium) respond to visible light.

Experimental Apparatus

The standard apparatus consists of an evacuated quartz-glass tube containing an emitter plate C (cathode) and a collector plate A (anode), illuminated through a quartz window by monochromatic light.

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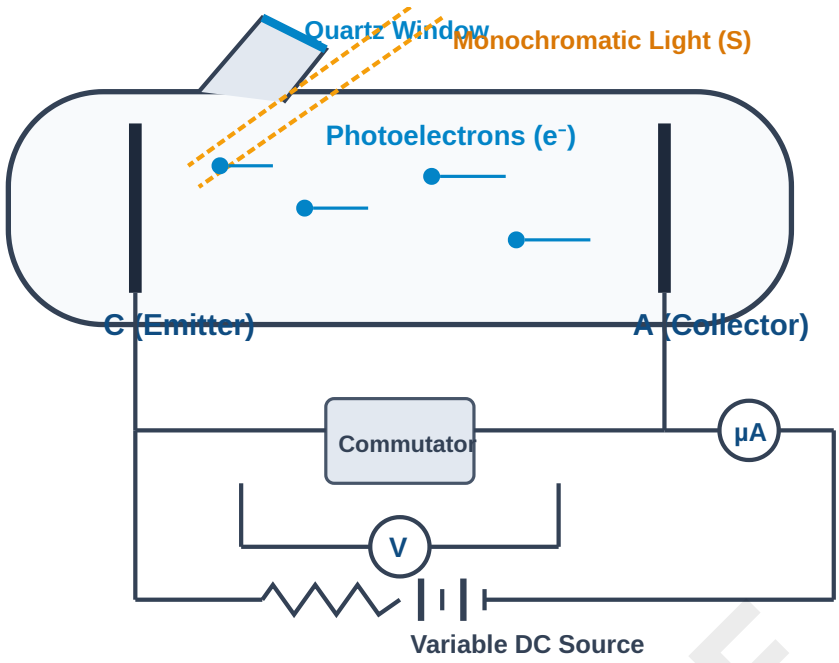


Figure 1: Schematic diagram of the experimental arrangement for studying the photoelectric effect.

5. Experimental Laws & Graphical Analysis

Systematic variation of radiation intensity, collector potential, and radiation frequency yielded four fundamental experimental results:

A. Effect of Radiation Intensity on Photocurrent

At fixed incident frequency ($\nu > \nu_0$) and fixed accelerating collector potential, the photoelectric current is **directly proportional to the intensity** of incident light.

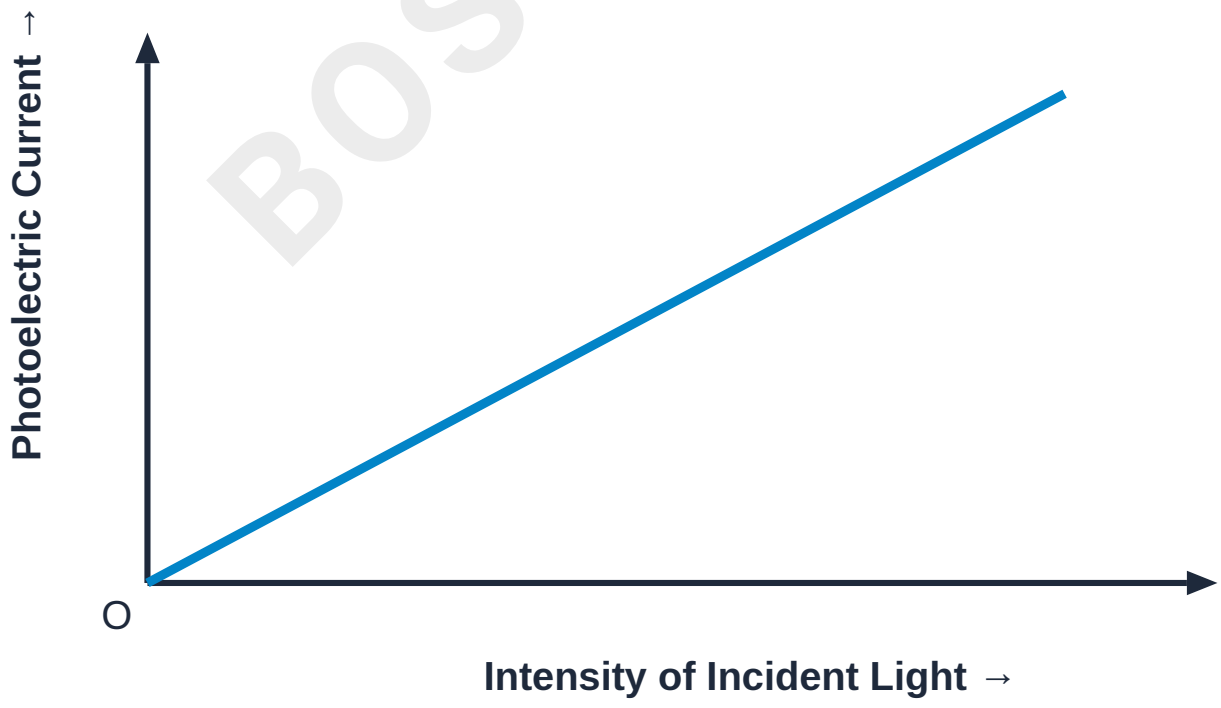


Figure 2: Linear dependence of photocurrent on light intensity at constant frequency ($\nu > \nu_0$).

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Physical significance: Higher intensity light delivers more energy packets per unit area per unit time, resulting in the ejection of a proportionally larger number of electrons per second.

B. Effect of Potential on Photocurrent (Different Intensities, Fixed Frequency)

When collector plate A is made positive relative to emitter C, the current rises with accelerating voltage until it reaches a plate-saturating limit known as the **saturation current**. At saturation, all electrons emitted per second are collected.

When collector plate A is given a negative (retarding) potential, electrons are repelled. The current decreases smoothly until it drops precisely to zero at a characteristic retarding voltage known as the **stopping potential (V_0)** or **cut-off potential**.

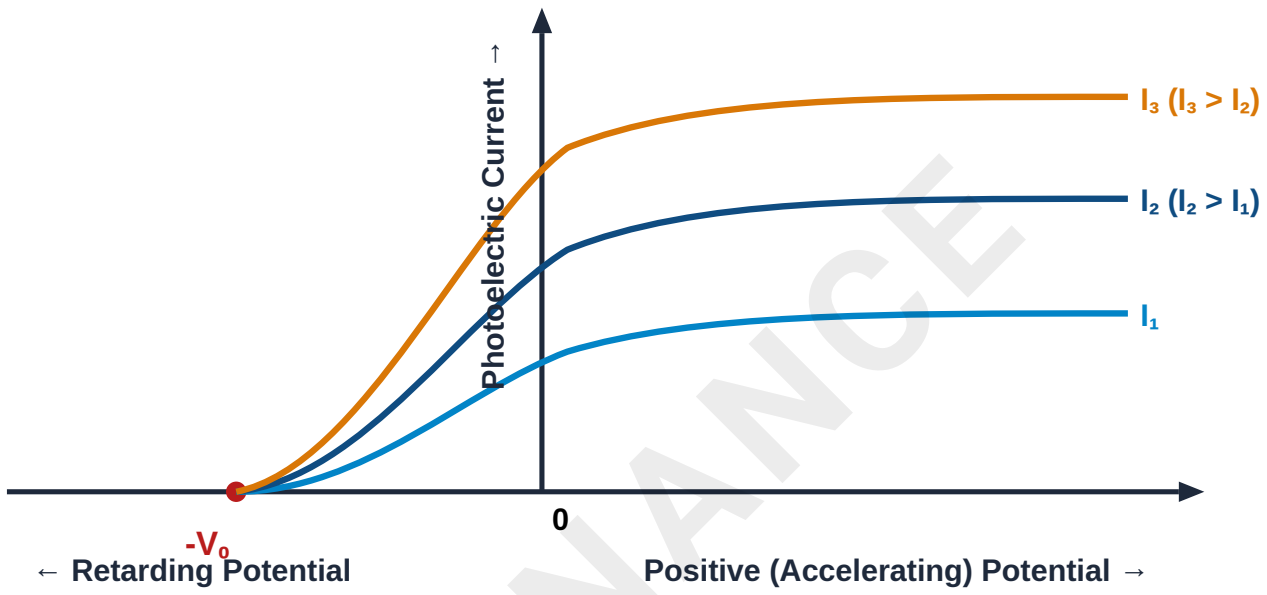


Figure 3: Current vs. Collector potential for different light intensities at constant frequency. The stopping potential $-V_0$ is identical for all intensities.

FUNDAMENTAL STOPPING POTENTIAL ENERGY RELATION

$$K_{\max} = \frac{1}{2}mv_{\max}^2 = eV_0$$

The stopping potential V_0 is completely independent of radiation intensity. Greater intensity increases the saturation current, but does not alter the maximum kinetic energy of the ejected electrons.

C. Effect of Incident Radiation Frequency on Stopping Potential

When the collector potential is varied for different frequencies $\nu_3 > \nu_2 > \nu_1$ at a fixed, uniform intensity, each frequency yields the *same saturation current* (since equal intensity at adjusted photon flux produces identical maximum collection rates), but *distinct stopping potentials*:

$$|V_{03}| > |V_{02}| > |V_{01}|$$

BOSONANCE

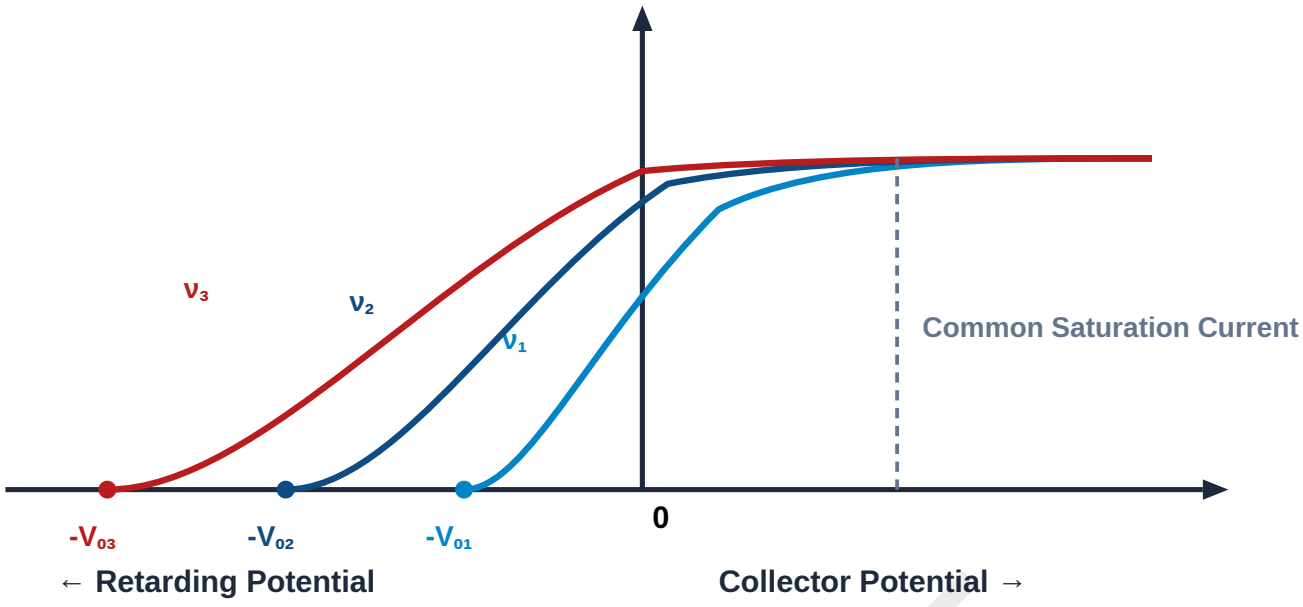


Figure 4: Variation of photocurrent with collector plate potential for different incident light frequencies ($\nu_3 > \nu_2 > \nu_1$) at identical saturation current.

D. Stopping Potential vs. Frequency Linear Relation

Plotting the stopping potential V_0 as a function of incident light frequency ν yields a straight line with a characteristic threshold frequency intercept on the horizontal axis:

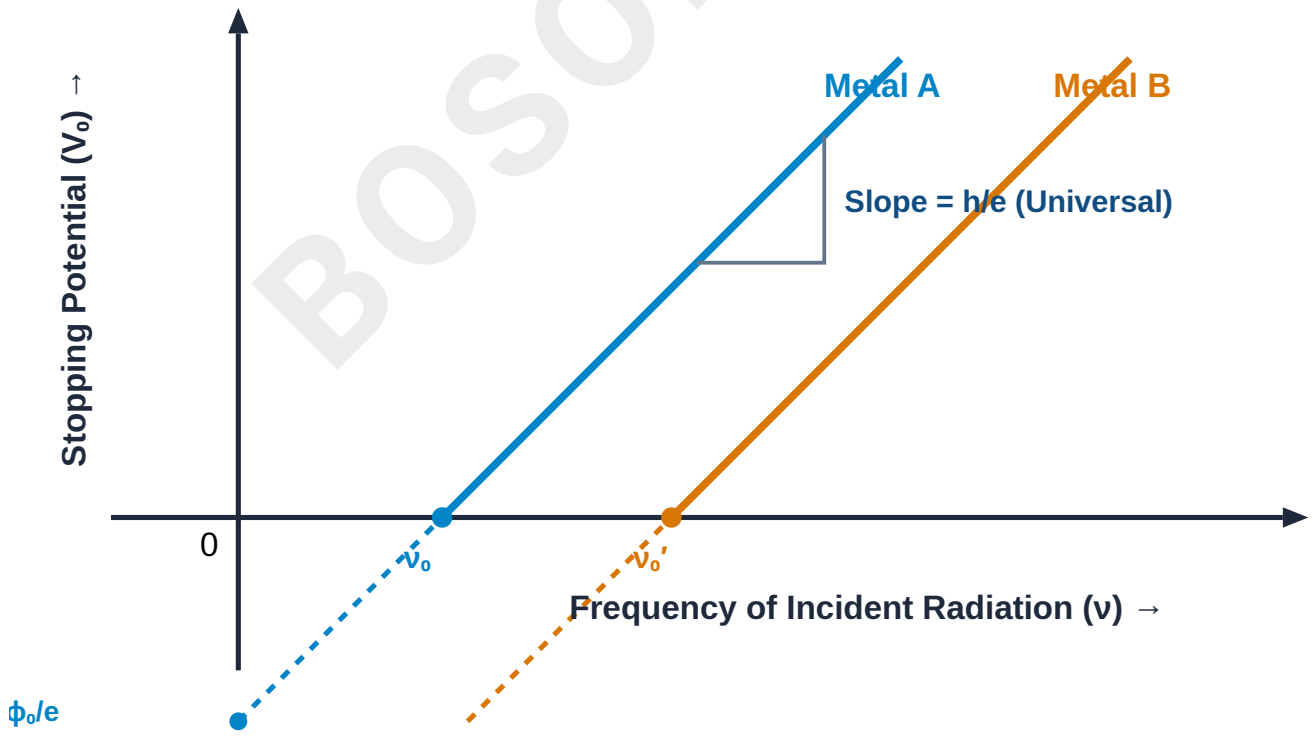


Figure 5: Stopping potential V_0 as a function of frequency ν. The lines are parallel because their slope (h/e) is a universal constant.

BOSONANCE

6. Classical Wave Theory Failure vs. Einstein's Photon Solution

Classical electromagnetic theory assumes that light is a continuous wave where radiant energy is distributed uniformly across the wavefront. This model fails completely when confronted with experimental observations:

Experimental Observation	Classical Wave Theory Prediction	Einstein's Quantum Solution
1. Dependence of $K_{\max}$ on Frequency, Not Intensity	Higher light intensity means larger electric field amplitude (E_0), so electrons should absorb more energy and emerge with higher kinetic energy.	Energy is packaged into discrete quanta ($E = h\nu$). One electron absorbs one photon. Greater intensity delivers *more* photons per second, not more energetic photons.
2. Existence of Threshold Frequency (ν_0)	Any frequency of light, if shone with sufficient intensity or over enough time, should deposit enough energy to liberate an electron.	Photoelectric emission requires $h\nu \geq \phi_0$. If $\nu < \nu_0$, individual photons lack the escape energy, making emission impossible regardless of intensity.
3. Instantaneous Emission ($\tau \leq 10^{-9}$ s)	Because energy spreads continuously over the wavefront, each atomic-sized cross-section absorbs only a tiny fraction of power. Calculations show a single electron would require *hours or days* to accumulate enough escape energy.	Energy transfer occurs in a single collision event between a photon and a conduction electron. The transfer is instantaneous ($\sim 10^{-9}$ s or less).

7. Einstein's Photoelectric Equation

In 1905, Albert Einstein applied Max Planck's quantum concept to electromagnetic radiation: light propagates as discrete localized packets of energy called **quanta** (later named **photons**).

By conservation of energy in a single photon-electron collision:

EINSTEIN'S PHOTOELECTRIC EQUATION

$$h\nu = \phi_0 + K_{\max}$$

$$K_{\max} = h\nu - \phi_0 = h(\nu - \nu_0) = hc\left(\frac{1}{\lambda} - \frac{1}{\lambda_0}\right)$$

Since stopping potential satisfies $K_{\max} = eV_0$, dividing by electronic charge e gives:

STOPPING POTENTIAL FORMULATION

$$V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi_0}{e} = \left(\frac{hc}{e}\right)\frac{1}{\lambda} - \frac{\phi_0}{e}$$

Analytical Conclusions for NEET:

- **Slope of V_0 vs. ν graph:**

$$\text{Slope} = \frac{h}{e} \approx 4.14 \times 10^{-15}\ \text{V} \cdot \text{s} \quad (\text{Universal constant, identical for all metals})$$

- **Horizontal Intercept:** Setting $V_0 = 0$ gives the threshold frequency:

$$\nu_0 = \frac{\phi_0}{h}$$

- **Vertical Intercept:** Extrapolating the straight line to the vertical axis ($\nu = 0$) gives:

$$\text{Intercept} = -\frac{\phi_0}{e}$$

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- **Slope of $K_{\max}$ vs. ν graph:** Equal to Planck's constant $h = 6.626 \times 10^{-34}\ \mathrm{J \cdot s}$.

Millikan's Experimental Verification (1916): R. A. Millikan conducted rigorous decade-long experiments on alkali metals to test Einstein's linear relation. By measuring the slope (h/e) and multiplying by the known charge of the electron e, Millikan determined Planck's constant h to high precision, independently confirming the quantum theory of radiation.

8. The Photon Theory of Radiation

In interactions with matter (such as photoelectric emission and Compton scattering of X-rays from electrons in 1924), electromagnetic radiation behaves as streams of particles called **photons**.

Characteristic Properties of Photons:

1. **Energy:** Each photon of frequency ν and wavelength λ carries energy:
$$E = h\nu = \frac{hc}{\lambda}$$
2. **Linear Momentum:** Photons carry directional momentum:
$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{h}{\lambda}$$
3. **Speed:** In free space (vacuum), all photons travel at the invariant speed of light:
$$c = 3.0 \times 10^{8}\ \mathrm{m/s}$$
regardless of their frequency, intensity, or the relative motion of the source.
4. **Rest Mass:** The rest mass of a photon is strictly zero: $m_0 = 0$.
5. **Electrical Neutrality:** Photons have zero net electrical charge and are never deflected by static electric or magnetic fields.
6. **Intensity & Flux:** The intensity of radiation is proportional to the number of photons crossing per unit area per second. Increasing intensity changes photon count, not individual photon energy.
$$\text{Number of photons emitted per second: } n = \frac{P}{E} = \frac{P\lambda}{hc}$$
7. **Conservation Laws:** In photon-particle collisions, total energy and total linear momentum are conserved. However, the *total number of photons may not be conserved*—a photon may be completely absorbed or created.

BOSONANCE

9. Wave Nature of Matter: de Broglie Hypothesis

In 1924, French physicist Louis Victor de Broglie proposed a unifying hypothesis: *Since nature displays profound symmetry, and radiation exhibits dual wave-particle properties, material particles (electrons, protons, atoms) must also exhibit wave characteristics under suitable conditions.*

THE DE BROGLIE RELATION

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Here, λ is the wave attribute (wavelength) and $p = mv$ is the classical particle attribute (momentum), bridged by Planck's constant h.

Alternative Formulations for Kinetic Energy & Potential

- **In terms of Kinetic Energy (K):** Since $p = \sqrt{2mK}$:
$$\lambda = \frac{h}{\sqrt{2mK}}$$
- **For a Charged Particle Accelerated through Potential Difference (V):** Since $K = qV$:
$$\lambda = \frac{h}{\sqrt{2mqV}}$$
- **Special Case: Electron accelerated from rest through V volts:** Substituting $m_e = 9.109 \times 10^{-31}\ \text{kg}$, $e = 1.602 \times 10^{-19}\ \text{C}$, and $h = 6.626 \times 10^{-34}\ \text{J}\cdot\text{s}$:
$$\lambda_e = \frac{1.227}{\sqrt{V}}\ \text{nm} = \frac{12.27}{\sqrt{V}}\ \text{Å}$$

Why Are Matter Waves Not Observed in Macroscopic Everyday Objects?

For a macroscopic object (e.g., a 0.12 kg ball moving at $20\ \text{m/s}$), its momentum is $p = 2.4\ \text{kg}\cdot\text{m/s}$. Its de Broglie wavelength is:

$$\lambda = \frac{6.63 \times 10^{-34}}{2.4} \approx 2.76 \times 10^{-34}\ \text{m}$$

This wavelength is roughly 10^{-19} times smaller than the diameter of a single atomic nucleus, making diffraction effects undetectable. Conversely, for a $100\ \text{eV}$ electron, $\lambda \approx 0.12\ \text{nm}$, which is on the scale of interatomic lattice spacings in crystals, producing measurable wave diffraction.

10. Physical Constants, Symbols, Dimensions & Units

Physical Quantity	Standard Symbol	Dimensional Formula	SI Unit	Key Defining Equation
Planck's Constant	h	$[ML^2T^{-1}]$	$\text{J}\cdot\text{s}$	$E = h\nu$
Work Function	ϕ_0	$[ML^2T^{-2}]$	J or eV	$\phi_0 = h\nu_0$
Stopping Potential	V_0	$[ML^2T^{-3}A^{-1}]$	V (Volt)	$eV_0 = K_{\max}$
Threshold Frequency	ν_0	$[T^{-1}]$	Hz or s^{-1}	$\nu_0 = \phi_0/h$
de Broglie Wavelength	λ	$[L]$	m	$\lambda = h/p$
Specific Charge of Electron	e/m	$[M^{-1}TA]$	$\text{C}\cdot\text{kg}^{-1}$	$e/m = 1.76 \times 10^{11}\ \text{C/kg}$

BOSONANCE

11. Step-by-Step Worked Examples

WORKED EXAMPLE 1: LASER POWER AND PHOTON EMISSION RATE

A helium-neon laser emits monochromatic light of wavelength $\lambda = 632.8\ \text{nm}$ with an output power of $9.42\ \text{mW}$.
(a) Find the energy and momentum of each photon in the light beam.
(b) Calculate how many photons per second are emitted by the source.

Solution:
(a) Energy and Momentum:

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\ \text{J} \cdot \text{s})(3.0 \times 10^{8}\ \text{m/s})}{632.8 \times 10^{-9}\ \text{m}} = 3.14 \times 10^{-19}\ \text{J}$$

In electron-volts:

$$E = \frac{3.14 \times 10^{-19}\ \text{J}}{1.602 \times 10^{-19}\ \text{J/eV}} \approx 1.96\ \text{eV}$$

Photon momentum p:

$$p = \frac{h}{\lambda} = \frac{6.63 \times 10^{-34}\ \text{J} \cdot \text{s}}{632.8 \times 10^{-9}\ \text{m}} = 1.05 \times 10^{-27}\ \text{kg} \cdot \text{m/s}$$

(b) Photon Emission Rate: If N photons are emitted per second, the total radiant power is $P = NE$:

$$N = \frac{P}{E} = \frac{9.42 \times 10^{-3}\ \text{W}}{3.14 \times 10^{-19}\ \text{J}} = 3.0 \times 10^{16}\ \text{photons/second}$$

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WORKED EXAMPLE 2: STOPPING POTENTIAL AND MAXIMUM SPEED

The work function of caesium metal is $2.14\ \text{eV}$. When light of frequency $6.0 \times 10^{14}\ \text{Hz}$ is incident on its surface, photoemission occurs. Determine:
(a) The maximum kinetic energy of the emitted photoelectrons.
(b) The stopping potential V_0.
(c) The maximum speed of the emitted photoelectrons.

Solution:
(a) Maximum Kinetic Energy: Energy of the incident photon:

$$E = h\nu = (6.63 \times 10^{-34}\ \text{J}\cdot\text{s})(6.0 \times 10^{14}\ \text{s}^{-1}) = 3.978 \times 10^{-19}\ \text{J}$$

Converting into eV:

$$E = \frac{3.978 \times 10^{-19}}{1.60 \times 10^{-19}} = 2.486\ \text{eV}$$

From Einstein's photoelectric equation:

$$K_{\max} = E - \phi_0 = 2.486\ \text{eV} - 2.14\ \text{eV} = 0.346\ \text{eV} \approx 0.35\ \text{eV}$$

In Joules: $K_{\max} = 0.346 \times 1.60 \times 10^{-19}\ \text{J} = 5.54 \times 10^{-20}\ \text{J}$.
(b) Stopping Potential: Since $eV_0 = K_{\max}$:

$$V_0 = \frac{K_{\max}}{e} = \frac{0.346\ \text{eV}}{e} = 0.35\ \text{V}$$

(c) Maximum Speed:

$$K_{\max} = \frac{1}{2}mv_{\max}^2 \implies v_{\max} = \sqrt{\frac{2K_{\max}}{m}}$$

$$v_{\max} = \sqrt{\frac{2 \times 5.54 \times 10^{-20}\ \text{J}}{9.11 \times 10^{-31}\ \text{kg}}} = \sqrt{1.216 \times 10^{11}} \approx 3.49 \times 10^{5}\ \text{m/s}$$

WORKED EXAMPLE 3: DE BROGLIE WAVELENGTH COMPARISON

Calculate the de Broglie wavelength associated with:
(a) An electron accelerated through a potential difference of $100\ \text{V}$.
(b) A dust particle of mass $1.0 \times 10^{-9}\ \text{kg}$ drifting at a speed of $2.2\ \text{m/s}$.

Solution:
(a) For the accelerated electron: Using the simplified relation for electrons:

$$\lambda = \frac{1.227}{\sqrt{V}}\ \text{nm} = \frac{1.227}{\sqrt{100}}\ \text{nm} = \frac{1.227}{10}\ \text{nm} = 0.1227\ \text{nm} \approx 1.23\ \text{Å}$$

This wavelength is of the order of interplanar spacings in crystal lattices, so electrons display detectable wave-diffraction patterns.
(b) For the macroscopic dust particle:

$$p = mv = (1.0 \times 10^{-9}\ \text{kg}) \times (2.2\ \text{m/s}) = 2.2 \times 10^{-9}\ \text{kg}\cdot\text{m/s}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}\ \text{J}\cdot\text{s}}{2.2 \times 10^{-9}\ \text{kg}\cdot\text{m/s}} = 3.01 \times 10^{-25}\ \text{m}$$

This is far smaller than any measurable physical dimension, confirming that macroscopic matter does not manifest observable wave properties.

BOSONANCE

12. Original NEET UG Practice Questions with Worked Solutions

Practice Question 1 (Single Correct Choice)

When the distance between a small point light source and a photosensitive cathode is halved, the saturation photoelectric current becomes:

A. Unchanged

B. Doubled

C. Four times its initial value

D. Halved

Correct Answer: C

Worked Solution: For a point source of radiant power P, intensity follows the inverse-square law:

$$I \propto \frac{1}{r^2}$$

When the distance r is reduced by half ($r' = r/2$), the intensity increases by a factor of 4:

$$I' = \frac{P}{4\pi(r/2)^2} = 4I$$

Since the saturation photocurrent is directly proportional to the incident radiation intensity, the photocurrent also increases to four times its initial value.

BOSONANCE

Practice Question 2 (Single Correct Choice)

The threshold wavelength for photoelectric emission from a metal surface is $600\ \text{nm}$. When radiation of wavelength $400\ \text{nm}$ is incident on the surface, the stopping potential is:
(Take $hc \approx 1240\ \text{eV} \cdot \text{nm}$)

A. 1.03 V

B. 2.07 V

C. 3.10 V

D. 0.52 V

Correct Answer: A

Worked Solution: Work function of the metal:

$$\phi_0 = \frac{hc}{\lambda_0} = \frac{1240\ \text{eV} \cdot \text{nm}}{600\ \text{nm}} \approx 2.067\ \text{eV}$$

Energy of the incident photon:

$$E = \frac{hc}{\lambda} = \frac{1240\ \text{eV} \cdot \text{nm}}{400\ \text{nm}} = 3.10\ \text{eV}$$

Maximum kinetic energy:

$$K_{\max} = E - \phi_0 = 3.10\ \text{eV} - 2.067\ \text{eV} = 1.033\ \text{eV}$$

Since $eV_0 = K_{\max}$, the stopping potential is $V_0 = 1.033\ \text{V} \approx 1.03\ \text{V}$.

Practice Question 3 (Graph / Slope Analysis)

In an experimental study of the photoelectric effect, the graph between the stopping potential V_0 and the incident frequency ν is plotted for two metals X and Y with work functions $\phi_X = 2.0\ \text{eV}$ and $\phi_Y = 4.5\ \text{eV}$. Which statement is correct?

A. The straight line for Y has a steeper slope than the line for X.

B. Both straight lines have the identical slope equal to h/e.

C. The line for X intersects the frequency axis to the right of the line for Y.

D. The slope of each line depends on the intensity of light used.

Correct Answer: B

Worked Solution: According to Einstein's equation:

$$V_0 = \left(\frac{h}{e}\right)\nu - \frac{\phi_0}{e}$$

Comparing with $y = mx + c$, the slope $m = h/e$ depends solely on Planck's constant h and elementary charge e. Hence, the slope is a universal constant, identical for all photosensitive metals. Metal Y has a larger work function, so its horizontal intercept ($\nu_0 = \phi_0/h$) lies further to the right.

BOSONANCE

Practice Question 4 (Ratio Calculation)

Monochromatic light of frequency ν ejects photoelectrons with maximum kinetic energy K_1 from a metal surface. If the frequency of incident radiation is doubled to 2ν, the new maximum kinetic energy K_2 satisfies:

A. $K_2 = 2K_1$

B. $K_2 < 2K_1$

C. $K_2 > 2K_1$

D. $K_2 = K_1 + h\nu/2$

Correct Answer: C

Worked Solution: From Einstein's photoelectric equation:

$$K_1 = h\nu - \phi_0 \implies h\nu = K_1 + \phi_0$$

For incident frequency 2ν:

$$K_2 = 2h\nu - \phi_0 = 2(K_1 + \phi_0) - \phi_0 = 2K_1 + \phi_0$$

Since work function $\phi_0 > 0$, it follows directly that:

$$K_2 = 2K_1 + \phi_0 > 2K_1$$

The new kinetic energy is more than twice the original.

BOSONANCE

Practice Question 5 (de Broglie Wavelength of Charged Particles)

A proton (p) and an alpha particle (α) are accelerated from rest through the same potential difference V. The ratio of their associated de Broglie wavelengths $\lambda_p : \lambda_\alpha$ is:

A. $2\sqrt{2} : 1$

B. $1 : 2\sqrt{2}$

C. $4 : 1$

D. $2 : 1$

Correct Answer: A

Worked Solution: The de Broglie wavelength of a particle of mass m and charge q accelerated through potential V is:

$$\lambda = \frac{h}{\sqrt{2mqV}} \implies \lambda \propto \frac{1}{\sqrt{mq}}$$

For an alpha particle relative to a proton:

$$m_\alpha \approx 4m_p \quad \text{and} \quad q_\alpha = 2q_p$$

Taking the ratio:

$$\frac{\lambda_p}{\lambda_\alpha} = \sqrt{\frac{m_\alpha q_\alpha}{m_p q_p}} = \sqrt{\frac{4m_p \times 2q_p}{m_p \times q_p}} = \sqrt{8} = 2\sqrt{2}$$

Hence, $\lambda_p : \lambda_\alpha = 2\sqrt{2} : 1$.

Practice Question 6 (Assertion-Reasoning)

Assertion (A): When monochromatic light of frequency $\nu > \nu_0$ falls on a photosensitive metal, all emitted photoelectrons do not possess the same kinetic energy.
Reason (R): Conduction electrons in a metal occupy an energy distribution below the Fermi level, and electrons emitted from deeper layers undergo collisions before reaching the surface.

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

C. (A) is true but (R) is false.

D. (A) is false but (R) is true.

Correct Answer: A

Worked Solution: The work function ϕ_0 represents the *minimum* energy required to free the most energetic electrons located right at the surface. Free electrons deeper inside the metal or those initially occupying lower energy levels within the conduction band require more energy to escape and lose variable amounts of kinetic energy via collisions. Hence, their emitted kinetic energies range continuously from 0 up to $K_{\max}$. Both statements are true, and (R) correctly explains (A).

BOSONANCE

Practice Question 7 (Two-Statement Evaluation)

Consider the following statements regarding photons:
Statement I: All photons of light of a given frequency ν have the same energy and the same momentum, regardless of the intensity of the beam.
Statement II: In every photon-particle collision, the total energy, total momentum, and total number of photons are strictly conserved.

A. Both Statement I and Statement II are correct.

B. Both Statement I and Statement II are incorrect.

C. Statement I is correct but Statement II is incorrect.

D. Statement I is incorrect but Statement II is correct.

Correct Answer: C

Worked Solution: Statement I is correct: Photon energy $E = h\nu$ and momentum $p = h\nu/c$ depend purely on frequency and are invariant with beam intensity.
Statement II is incorrect: While total energy and total linear momentum are always conserved in any collision, the *number of photons is not necessarily conserved*; a photon may be absorbed completely (as in the photoelectric effect) or new photons may be created.

Practice Question 8 (Numerical Problem on de Broglie Wavelength)

The momentum of a particle is increased by 100%. The percentage decrease in its associated de Broglie wavelength is:

A. 100%

B. 50%

C. 25%

D. 75%

Correct Answer: B

Worked Solution: Initial momentum = p_1.
If momentum is increased by 100%, new momentum $p_2 = p_1 + 1.00p_1 = 2p_1$.
Since $\lambda = h/p$:

$$\lambda_2 = \frac{h}{p_2} = \frac{h}{2p_1} = \frac{\lambda_1}{2}$$

Percentage decrease in wavelength:

$$\frac{\lambda_1 - \lambda_2}{\lambda_1} \times 100\% = \frac{\lambda_1 - 0.5\lambda_1}{\lambda_1} \times 100\% = 50\%$$

BOSONANCE

13. Common Traps & High-Yield Revision Checklist

NEET Trap Warnings:

- **Confusing Intensity with Frequency:**
  - Increasing *intensity* increases the number of photons per second, thus increasing the *saturation current*. It has **zero effect** on stopping potential (V_0) or maximum kinetic energy ($K_{\max}$).
  - Increasing *frequency* increases photon energy, increasing $K_{\max}$ and stopping potential V_0. It does **not** increase saturation current if intensity is kept constant.
- **Sub-threshold Illumination:** If $\nu < \nu_0$ (or $\lambda > \lambda_0$), photoelectric emission is strictly *zero*, even if an ultra-intense laser is shone for hours.
- **Slope Invariance:** The slope of the V_0 vs. ν graph is always $\frac{h}{e}$ (universal). It does not change with different metals or differing surface treatments. Different metals only change the horizontal intercept (ν_0) and vertical intercept ($-\phi_0/e$).
- **Photon Conservation:** Remember that photons are not conserved particles. They can be created or annihilated during physical interactions.

Rapid Revision Checklist

- ☑ $1\ \text{eV} = 1.602 \times 10^{-19}\ \text{J}$
- ☑ Specific charge of electron: $e/m = 1.76 \times 10^{11}\ \text{C/kg}$
- ☑ Einstein's equation: $K_{\max} = eV_0 = h\nu - \phi_0 = hc/\lambda - \phi_0$
- ☑ Threshold frequency: $\nu_0 = \phi_0/h$; Threshold wavelength: $\lambda_0 = hc/\phi_0$
- ☑ Photon momentum: $p = E/c = h/\lambda$
- ☑ de Broglie wavelength: $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mK}} = \frac{h}{\sqrt{2mqV}}$
- ☑ Accelerated electron shortcut: $\lambda_e = \frac{1.227}{\sqrt{V}}\ \text{nm} = \frac{12.27}{\sqrt{V}}\ \text{Å}$
- ☑ Photoelectric time lag: $\leq 10^{-9}$ s (instantaneous)

14. Chapter Recap

The discovery of the electron and experimental investigations into the photoelectric effect revealed the fundamental limits of classical wave mechanics. Light interacts with matter not as a continuous wave, but as discrete quanta called **photons**, each transferring energy $E = h\nu$ and momentum $p = h/\lambda$ in single instantaneous collisions. Einstein's photoelectric equation $eV_0 = h\nu - \phi_0$ perfectly explains the existence of threshold frequency, the linear dependence of stopping potential on frequency, the independence of $K_{\max}$ from light intensity, and the instantaneous nature of emission. In complement, de Broglie's hypothesis revealed that material particles also possess wave-like properties with wavelength $\lambda = h/p$. This wave nature is physically significant for subatomic particles (such as electrons) while being imperceptible for macroscopic bodies, cementing the dual nature of radiation and matter as a cornerstone of modern quantum physics.

BOSONANCE